The empirical recurrence for OEIS A281464, proved by transfer matrix

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Abstract

OEIS A281464 counts $n \times 3$ arrays over $\{0, \ldots, 1\}$ subject to a condition imposed on every CELL over a neighbour set the entry names, and counted UP TO RELABELLING of the alphabet; it carries an empirical recurrence of order 7 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. What is counted is equality patterns, which is not a walk count — but it is a fixed rational combination of walk counts, one for each alphabet size, and that combination is again C-finite. Whether the conjectured recurrence annihilates it is then decided by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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1 The conjecture

OEIS A281464 is "Number of n X 3 0..1 arrays with no element equal to more than two of its horizontal, diagonal or antidiagonal neighbors and with new values introduced in order 0 sequentially upwards.". It has offset 1 and begins

$$4, 17, 24, 60, 133, 283, 634, 1419, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 2*a(n-1) + 3*a(n-3) - 3*a(n-4) - 4*a(n-5) + 4*a(n-6) - 4*a(n-7) for n>9.

As of the "Last modified" line on the live entry (Feb 20 2019 05:30:08, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

2 Patterns are a combination of counts

The clause "new values 0..1 introduced in row major order" says that the arrays counted are the canonical representatives of their EQUALITY PATTERNS: two arrays that differ only by a permutation of the alphabet are one object. Write N_j for the number of admissible patterns using exactly j letters and L_i for the number of admissible ARRAYS over an alphabet of i letters. An array over i letters is a pattern together with an injection of its letters into the alphabet, so

$$L_i \ = \ \sum_j N_j \, i(i-1)\cdots(i-j+1),$$

a triangular system with nonzero diagonal, hence invertible over $\mathbb{Q}$. The entry counts $a = N_1 + \cdots + N_2$, so

$$a \ = \ \left(\mathbf{1}^\top F^{-1}\right) L,$$

a FIXED rational combination of the L_i, where F is the matrix of falling factorials. Each L_i is a walk count on its own transfer graph, so a is a walk count on the disjoint union of those graphs with the combination's coefficients placed in the starting vector; the coefficients are rational and some are negative, which costs nothing in what follows.

This step needs the condition itself to be unchanged by permuting the alphabet, and it is: the condition below is stated through equality between entries and never names a particular value or compares sizes.

3 Cells, not blocks

The entry names the neighbours it means — horizontal is left and right, vertical is up and down, diagonal is northwest-to-southeast, antidiagonal is northeast-to-southwest, king-move is all eight — and every such set lies inside the 3×3 square centred on the cell. Here the set is $\{(0,-1),(0,1),(-1,-1),(1,1),(-1,1),(1,-1)\}$, written as offsets (row, column) from the cell. Every cell carries the condition

> no cell is equal to more than 2 of its horizontal, diagonal or antidiagonal neighbours,

and the count of neighbours enters it, so the boundary is not a detail that can be papered over: a cell in the interior has 6 neighbours in this set and a cell on the boundary fewer, and a condition such as "a strict majority of its neighbours" therefore means different things at the two. In particular one cannot pad the array with a border of zeros — that would give every cell the full 6 and change what is being counted. The outside is therefore carried explicitly, as a sentinel row above the first and below the last and as a sentinel column either side, and a neighbour that falls outside is simply absent from both counts.

Write an array as a sequence of rows $u_1, \dots, u_L$, each in $\{0, \dots, 1\}^3$. The neighbourhood of a cell in row u_i meets u_{i-1}, u_i and u_{i+1} and nothing else. Take as vertices the pairs $(r, s) \in (\{0, \dots, 1\}^3 \cup \{\varnothing\}) \times \{0, \dots, 1\}^3$, and put an edge $(r, s) \to (s, t)$ when row s satisfies the condition at every one of its 3 cells, given r above and t below. An array with L rows is then a walk of length $L-1$ starting at $(\varnothing, u_1)$, each step settling one row, and the last row is settled by the terminal weight, which evaluates it with $\varnothing$ below. Hence, with M the adjacency matrix, ι the starting vector and τ the terminal weight,

$$a(n) \;=\; \frac{1}{2}\, \iota^{\top} M^{\,n-1} \tau,$$

the exponent fixed by the entry's own shape and checked against its published terms. In particular a is C-finite.

4 The proof

Lemma 1. *Let $q(t) = t^7 - \sum_i c_i t^{7-i}$ be the characteristic polynomial of the conjectured recurrence. Then the residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^{\top} M^{\,j} q(M) \tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^{\top} M^j q(M) \tau = 0$ for all large j; and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+7} - \sum_i c_i M^{j+7-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^{\top} M^j q(M) \tau$ satisfy the monic linear recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$$a(n) \;=\; 2\,a(n-1) + 3\,a(n-3) - 3\,a(n-4) - 4\,a(n-5) + 4\,a(n-6) - 4\,a(n-7)$$

for every $n > 9$.

Proof. The vector $w = q(M)\tau$ was formed by 7 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots$, again exactly. Those integers vanish from the index corresponding to $n = 9 + 1$ onwards, and the run was continued until the working vector was identically zero, which settles all larger j at once. By Lemma 1 the recurrence holds for every $n > 9$. The bound is exact: at $n = 9$ the recurrence fails on the entry's own published terms, so no larger range holds. □

5 Verification

Three checks, each independent of the algebra above.

First, the transfer model reproduces all 29 terms the entry publishes, exactly, in integer arithmetic. That check earned its place here. The neighbour offsets are named in the array's own frame — horizontal means along a row — but when the entry fixes the first dimension, as in "$2 \times n$", the walk runs along columns and the two components of every offset swap. Sets such as king-move, or horizontal together with vertical, are unchanged by that swap and hide the error completely; a set such as horizontal, diagonal and antidiagonal is not, and the counts came out as those of a different member of the same family until the offsets were transposed.

Second, the conjectured recurrence was evaluated directly on the entry's own published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A281464.

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